

Stochastic Optimal Energy Storage Management for Energy Routers Via Compressive Sensing

Hong Liang , Haochen Hua , Member, IEEE, Yuchao Qin, Maojiao Ye, Member, IEEE, Shuqing Zhang , Member, IEEE, and Junwei Cao , Senior Member, IEEE

Abstract—The functionality of energy routing among microgrids is becoming increasingly important with the progress of deploying smart power systems all over the world. For higher energy routing performance and better renewable energy integration, a new type of electrical device, called energy router (ER), is being developed as a part of the infrastructure of the future energy Internet (EI). Generally, the long-term operation of ERs requires an effective energy management scheme for the energy storage inside these devices. In this article, considering the randomness of power generation by renewable energy sources and the stochastic power usage of loads in EI scenario, the compressive sensing is adopted for the solution to the nonlinear energy storage management problem which is essential for the design of ERs. The compressive sensing method used in this article is proven to be more efficient than the conventional Monte–Carlo methods and polynomial chaos expansion method, and the performance of the proposed method is evaluated with numerical examples.

Index Terms—Compressive sensing, energy Internet, energy router, energy storage, stochastic optimization.

NOMENCLATURE

EI	Energy Internet.
ER	Energy router.
MG	Microgrid.
DG	Distributed controllable power generator.
RES	Renewable energy source.

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Hong Liang, Shuqing Zhang, and Junwei Cao are with Tsinghua University, Beijing 100084, China (e-mail: lianghong@tsinghua.edu.cn; zsq@tsinghua.edu.cn; jcao@tsinghua.edu.cn).

Haochen Hua is with Hohai University, Nanjing 210098, China (e-mail: huahc16@tsinghua.org.cn).

Yuchao Qin is with the University of Cambridge, CB2 1TN Cambridge, U.K. (e-mail: qinyc17@mails.tsinghua.edu.cn).

Maojiao Ye is with the School of Automation, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: mjye@njut.edu.cn).

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ΔP_{Load}	Power deviation of loads.
ΔP_{RES}	Power deviation of RESs.
ΔP_{DG}	Power deviation of DGs.
ΔP_{out}	Aggregate power transmission request from external ERs.
ΔP	Power input/output of the energy storage in ER.
S_{ER}	Energy storage status of the considered ER.
CS	Compressive sensing.
MC	Monte–Carlo.
PCE	Polynomial chaos expansion.

I. INTRODUCTION

THE energy Internet (EI) has changed the mode and the structure of energy supply and demand in the sense that the centralized top-down energy management mode is being transformed into a combination of an interactive centralized top-down mode and a distributed bottom-up mode, such that consumers are satisfied with the demand for power services, the contradiction between power peaks and valleys can be effectively solved, and the requirements of energy security and reliability can be met [1].

Various energy forms, such as cooling, heating, electricity, and gas, are fully integrated in the generation, transmission, and consumption side, making full use of the complementary characteristics of multiple energy sources, and increasing the energy utilization efficiency of the terminal through energy cascade utilization. Through the free choice and interactive management of energy sources, the energy industry has also been influenced by the Internet; related industries have been risen, and the economic structure and even social habits are being reformed [2].

In future EI scenarios, demand-side response based on big data and wide-area information platform would improve the efficiency of energy production and transmission [3]. Located at the connection points of EI, ERs are the key equipment to realize not only information transmission but also open and peer-to-peer transmission routing of energy [4]. The energy router (ER; also called energy hub, or electric router), an emerging device based on advanced power electronic techniques, shall be able to implement flexible and dynamic power distribution in EI analogous to the character of information routers in the Internet. With the support of ER, communication network, and power

network, the bottom-up infrastructure in the future EI can be built [5].

In recent years, the investigation of ER regarding different aspects has been popular. The architectures, functionalities, and demonstration of ER have been introduced in [6]. Similar to the data caching function of routers in the Internet, ER needs the capability of fast and reliable energy storage to enable the flexible energy sharing in future EI scenarios. Although research regarding ER has been popular during the past five years, only prototypes of ERs are developed in laboratories, and there has been no commercial products so far. In this article, the energy storage management for ERs is considered in a macro perspective, in which sense, control strategies shall be designed for all microgrids (MGs) in EI, rather than one single ER in [7]. In particular, the stochastic feature of the power generation by renewable energy sources (RESs) and the power of loads are considered. We assume the controllers are set in distributed controllable power generators (DGs) and ERs. Both ordinary and stochastic differential equations are adopted to describe the power dynamics of the considered system. Then, the energy storage management task for ER is formulated as a nonlinear stochastic optimization problem.

It is almost impossible to solve this problem analytically, and numerical methods shall be considered. To address this issue, adaptive dynamic programming (ADP), proposed by Werbos [8], has widely been applied to obtain the optimal energy management policies in EI situation. For example in [9], it proposes an ADP-based approach for the economic dispatch of MG with DGs and a modified value function update strategy is included in this work. A lookup tables-based ADP algorithm for the real-time energy management of the MG under uncertainties is proposed in [10], while the optimal operation of the MG is also formulated as a stochastic nonlinear programming problem. In [11], they use the ADP method to solve a smart home energy management system and improve the performance of such problem by considering uncertainties of the system. A stochastic gas-power network constrained unit commitment model considering both combined-cycle units and gas network is established in [12]; in order to prevent the curse of dimensionality, they also use ADP method to deal with the stochastic control problem. An action-dependent heuristic dynamic programming (ADHDP) method is applied to the optimal control of dynamic energy management systems for MGs [13]. Notably, ADHDP method is also denoted as Q-learning. Moreover, due to the rapid development of artificial intelligence, reinforcement learning method becomes more attractive in solving model-free or model-based problems of EI, such as in [14], an artificial neural network-based reinforcement learning algorithm was proposed to manage an optimal multienergy management-based energy routing design problem. In [15] concentrates on a model-free energy routing strategy problem, and the actor-critic reinforcement learning approach is applied to solve the optimal control problem. A multiagent-based model is used to study distributed energy management in an MG in [16], and a reinforcement learning algorithm was developed to derive the optimal strategies for energy management and load scheduling without prior information of the MG system.

An alternative approach is to combine Monte-Carlo (MC) method with heuristic optimization algorithm [17] or traditional iterative method such as gradient decent by Newton et al. Normally, the MC method has a relatively low convergence rate. Thus, in order to achieve a desirable computation accuracy, it often results in a prohibitively high computational complexity. Hence, aiming to overcome the low efficiency caused by the huge amount of computation, more efficient methods for solving stochastic equations should be applied, and hence instead of using MC, an alternative method is polynomial chaos expansion (PCE) [18], which has been proved efficient in solving stochastic equations, but unlike the MC method, it usually requires recoding the system into a much higher dimensional problem which makes it complex in programming. However, due to the sparsity of the solution for stochastic system, compressive sensing (CS) method which is emerged from the field of sparse signal recovery [19] can be used to overcome the disadvantages of both PCE and MC methods, because it does not require recoding the system like PCE method, nor does it require a large number of samples like MC method. As a whole, the novelty aspects of this article are summarized as follows.

1) A novel gradient-based method with CS has been proposed to solve high-dimensional stochastic optimization problem. The main innovation of our proposed algorithm is to apply CS method instead of MC method to deal with stochastic system in the iteration of traditional optimization algorithm, and it has been shown effective.

2) In this article, an energy storage management problem for the ER is formulated as an optimal control problem, taking the ER system's nonlinearity and stochasticity into consideration, which has not been fully investigated in previous works.

3) Simulation results show that when solving the aforementioned problem, CS method converges significantly faster than MC method, and is much more easy to be realized by programming than PCE method, which indicates that CS can save computational cost dramatically when applied in such kind of stochastic optimization problem. It is notable that our proposed algorithm can be used not only in the problem of ER energy management, but also in a variety of fields where similar optimal control problems are expected to be solved.

The rest of this article is organized as follows. System modeling and problem formulation are provided in Sections II and III, respectively. The solution to the considered optimization problem is investigated in Section IV. Section V provides some simulations. Finally, Section VI concludes this article.

II. SYSTEM MODELING

In this article, we consider a special EI scenario where a variety of MGs are interconnected without access to the utility grid, in the sense that such EI is functioning in an OFF-grid mode. A simplified configuration is illustrated in Fig. 1. Such EI scenario, in which the main power generation relies on RESs, is suitable for some rural or remote areas; see, e.g., [20].

For the EI scenario in Fig. 1, our focus is put on one of the interconnected MGs. Each MG consists of multiple types of RESs, DGs, and loads which are all connected with an ER,

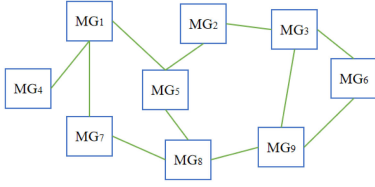


Fig. 1. Typical off-grid EI network.

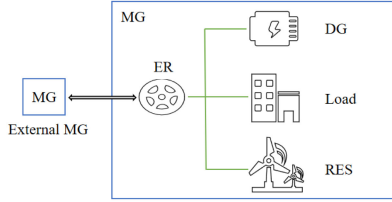


Fig. 2. A typical MG.

and such ER is connected with an external ER in the other MG; see Fig. 2. The linearized modeling approach is adopted to describe the power dynamics of each component in MG. To reveal the uncertainty and variation of power of load and RESs [21], stochastic differential equations driven by scalar Brownian motion are given as follows:

$$d\Delta P_{Load} = -\theta_l(\Delta P_{Load} - \mu_l)dt + \sigma_l dW_l \quad (1)$$

$$d\Delta P_{RES} = -\theta_r(\Delta P_{RES} - \mu_r)dt + \sigma_r dW_r \quad (2)$$

where W_l and W_r refer to scalar Brownian motions, and scalars θ_l , θ_r , μ_l , μ_r , σ_l , and σ_r are system parameters that can be measured by parameter estimation methods. For notation simplicity, time t of all variables throughout this context is omitted.

The DGs in our considered MG can be microturbines, diesel engine generators, fuel cells, etc. The following ordinary differential equation shows the generalized power dynamics of DGs.

$$d\Delta P_{DG} = -\frac{1}{T_{DG}}(\Delta P_{DG} - u_G)dt \quad (3)$$

where T_{DG} stands for the time constant of DGs, u_G is the desired power adjustment for the DGs, which can be viewed as control input of DGs.

Power generation by RESs and power usage caused by loads flow into and out of the ER in Fig. 2, respectively. Thus, it is obvious that the power to be stored in ER is

$$\Delta P = \Delta P_{RES} + \Delta P_G - \Delta P_{Load} + \Delta P_{out}(1 - u_{ER}) \quad (4)$$

where u_{ER} denotes the average transmission rejection rate for other ERs connected to the considered ER. In this sense, u_{ER} is the control input for the investigated ER. For such ER, since the total power required from the accessed external ERs is also random, depending on various complex conditions, stochastic differential equation similar with (1) and (2) are used to describe ΔP_{out} . Let us denote W_p as the stochastic process in the same probability space of W_l and W_r . Then, we have

$$d\Delta P_{out} = -\theta_p(\Delta P_{out} - \mu_p)dt + \sigma_p dW_p \quad (5)$$

where scalars θ_p , μ_p , and σ_p are system parameters.

The energy storage status of the considered ER, denoted as S_{ER} , is actually the state of charge of ER's energy storage component, and it is represented as

$$dS_{ER} = \eta \Delta P dt \quad (6)$$

where η is a constant related to the energy efficiency and capacity of the energy storage attached in the ER, and the value of S_{ER} is naturally defined to be within $[0, 100\%]$.

In this sense, the linearized models from (1) to (6) fully describe the dynamics of one typical MG in the considered EI scenario. For similar modeling approaches, readers can refer to [22].

Next, let $x = [\Delta P_{Load}, \Delta P_{RES}, \Delta P_G, \Delta P_{out}, S_{ER}]'$, $u = [u_G, u_{ER}]'$, and $W = [W_l, W_r, W_p]'$. The system from (1) to (6) can be rewritten as

$$dx = [A(u)x + Bu + C]dt + DdW \quad (7)$$

where matrices

$$A(u) = \begin{pmatrix} -\theta_l & 0 & 0 & 0 & 0 \\ 0 & -\theta_r & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{T_g} & 0 & 0 \\ 0 & 0 & 0 & -\theta_p & 0 \\ -\eta & \eta & \eta & \eta(1 - u_{ER}) & 0 \end{pmatrix} \quad (8)$$

$$B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ \frac{1}{T_g} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} \theta_l \mu_l \\ \theta_r \mu_r \\ 0 \\ \theta_p \mu_p \\ 0 \end{pmatrix}, \quad D = \begin{pmatrix} \sigma_l & 0 & 0 \\ 0 & \sigma_r & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma_p \\ 0 & 0 & 0 \end{pmatrix}. \quad (9)$$

In this sense, our studied MG power dynamics have been transformed into a stochastic control system (7), where $x(t)$ is the system state, and $u(t)$ is the system control input.

It is notable that matrix $A(u)$ is not a time-invariant parameter matrix. Essentially, it is a function of $u_{ER}(t)$. The term $A(u)x$ indicates that system (7) is indeed a *nonlinear* system.

III. PROBLEM FORMULATION

In this section, the energy storage management for the considered ER is formulated as an optimal control problem.

Due to the energy management principle of EI, the autonomous operation of the power system shall be preferably achieved at the MG level [5]. As key devices providing the energy-sharing functionality among MGs, ERs are expected to be of the ability to achieve long-term operations with high stability. To this end, one of the important aspects is to maintain the energy storage status of the considered ER at a stable level over time [7]. This target could be achieved by restricting the power throughput of the energy storage in the considered ER. However, overrestricted power throughput may impair the energy storage capability of the ER and reduce the stability of the entire EI system. In this sense, instead of utilizing regular quadratic cost functions, the objective function in (10) is adopted in this article, whose value should be minimized by designing proper control

strategies.

$$\mathcal{J}_0 = \mathbb{E} \int_0^T \left\{ \varepsilon_1 \log(1 + e^{\Delta P^2}) + \varepsilon_2 u_G^2 \right\} dt \quad (10)$$

where $\mathbb{E}$ is the mathematical expectation, and ε_1 and ε_2 are weight coefficients. The interpretation of each term in (10) is as follows.

For stochastic control problems, the traditional LQ problem only considers the mean of the cost and ignores the higher order momenta. In order to incorporate higher order momenta of the objective function, we made a little modification by adopting the term $\log(1 + e^{\Delta P^2})$ in (10); the objective function (10) can consider the higher order momenta of the stochastic inputs according to the design of the exponential-quadratic control objective function in risk-sensitive control theory [23], and the energy storage capability of ERs could be better utilized compared to simply adopting ΔP^2 in (10). Given the same value of ΔP , for instance, for a small ΔP , we have $\log(1 + e^{\Delta P^2}) \geq \Delta P^2$. This means that the optimal controller for (10) would tend to address small power deviation ΔP via the adjustment of DGs. Additionally, the second term in the integral of (10) is used to avoid the overcontrol of DGs.

Let $C_{\min}$ and $C_{\max}$ be the minimum and maximum values of S_{ER} , and $\Delta P_{\text{Load}}^{\max}$, $\Delta P_{\text{RES}}^{\max}$, and $\Delta P_G^{\max}$ be the maximum power variation constraints of ΔP_{Load} , ΔP_{RES} , and ΔP_G , respectively. The optimal control problem is then formulated as (11).

$$\begin{aligned} \min_{u_G, u_{ER}} \quad & \mathcal{J}_0 \\ \text{s.t.} \quad & (1) - (6) \\ & \Delta P_{\text{Load}} \leq |\Delta P_{\text{Load}}^{\max}| \\ & \Delta P_{\text{RES}} \leq |\Delta P_{\text{RES}}^{\max}| \\ & \Delta P_G \leq |\Delta P_G^{\max}| \\ & C_{\min} \leq S_{ER} \leq C_{\max} \\ & 0 \leq u_{ER} \leq 1. \end{aligned} \quad (11)$$

In order to deal with the inequality constraints in (11), two penalty terms $\mathcal{P}_1$ and $\mathcal{P}_2$ are introduced to the objective function. Specifically, the penalty term $\mathcal{P}_1$ for S_{ER} and u_{ER} is designed in the following form to ensure S_{ER} and u_{ER} vary near the midpoints of the constraints as much as possible.

$$\mathcal{P}_1 = \left(\frac{S_{ER} - \frac{1}{2}(C_{\max} + C_{\min})}{C_{\max} - C_{\min}} \right)^2 + \left(u_{ER} - \frac{1}{2} \right)^2 \quad (12)$$

Moreover, to keep the power variations within the maximum constraints, the penalty term $\mathcal{P}_2$ for ΔP_{Load} , ΔP_{RES} , and ΔP_G has the following form:

$$\mathcal{P}_2 = 10^5 (\mathbb{I}_{\text{Load}} + \mathbb{I}_{\text{RES}} + \mathbb{I}_G) \quad (13)$$

where $\mathbb{I}_{\text{Load}}$ is the characteristic function defined by

$$\mathbb{I}_{\text{Load}} = \begin{cases} 0 & \text{if } \Delta P_{\text{Load}} \leq |\Delta P_{\text{Load}}^{\max}| \\ 1 & \text{otherwise} \end{cases}. \quad (14)$$

Similarly, $\mathbb{I}_{\text{RES}}$ and $\mathbb{I}_G$ are defined in the same way. Multiplying the characteristic function by 10^5 can prevent the power variations from exceeding the limits.

With a large weight coefficient for the penalty term $\mathcal{P} = \mathcal{P}_1 + \mathcal{P}_2$, the optimal controller would presumably keep S_{ER} and u_{ER} around the midpoints of their allowed ranges and control the power variations within reasonable boundaries. In this manner, the optimal control problem with inequality constraints in (11) is transformed to be a relaxed problem (15).

$$\begin{aligned} \min_{u_G, u_{ER}} \quad & \mathcal{J} = \mathbb{E} \int_0^T \left\{ \varepsilon_1 \log(1 + e^{x'Qx}) + \varepsilon_2 u'Ru + \varepsilon_3 \mathcal{P} \right\} dt \\ \text{s.t.} \quad & (1) - (6) \end{aligned} \quad (15)$$

where ε_3 is the weight coefficients

$$Q(u) = \begin{pmatrix} 1 & -1 & -1 & u_{ER} - 1 & 0 \\ -1 & 1 & 1 & (1 - u_{ER}) & 0 \\ -1 & 1 & 1 & (1 - u_{ER}) & 0 \\ u_{ER} - 1 & 1 - u_{ER} & 1 - u_{ER} & (1 - u_{ER})^2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (16)$$

$$R = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (17)$$

By solving the stochastic nonlinear optimal control problem (15), the optimal energy storage management strategy for the considered ER could be obtained. As mentioned in Section I, the conventional methods, such as dynamic programming or the MC method, are limited in computational efficiency and may not be feasible for high-dimensional problems. Since the polynomial expansion of stochastic process usually has sparsity [24], the CS method which has been developed rapidly in recent years is used to solve the aforementioned control problem. The detailed methodologies are given in the next section.

IV. SOLVING THE OPTIMIZATION PROBLEM

In this section, an algorithm based on CS has been proposed to solve the optimization problem (15). First of all, we review the process of solving differential equations using the CS method. Then, by combining the CS with gradient descent method, we propose a new iterative algorithm, named as CS-GDM, to solve the optimization problem (15).

A. Compressive Sensing for Differential Equations

The process of using CS method to solve stochastic differential equations can be summarized as three steps:

- s1) Representation of Brownian motion via Fourier expansion;
- s2) PCE;
- s3) Sparse recovery using CS method.

For step s1, choosing an orthonormal basis $\{\varphi_i\}_{i=1}^{\infty}$, we have

$$W(s) = \sum_{i=1}^{\infty} \xi_i \int_0^s \varphi_i(\tau) d\tau \quad (18)$$

where $W(s)$ denotes Brownian motion, $\{\xi_i\}_{i=1}^{\infty}$ are independent identically distributed random variables selected from the standard normal distribution, and expansion (18) uniformly converges to $W(s)$ in the mean square sense.

Expanding the Brownian motion is to discretize the original problem. Let $i \in [1, d]$, we truncate the expansion with d terms in (18). Then, the Brownian motion in system (7) can be expressed via $\xi = \{\xi_i\}_{i=1}^d$, so the system state $x(t)$ is actually related to ξ . More specifically, $x(t)$ should be written as $x(t, \xi)$, then we can use a set of stochastic polynomials to expand the system state $x(t, \xi)$, which is described as step s2.

In this article, since the random variables ξ comes from the standard normal distribution, we choose the generalized Hermite polynomial basis as the stochastic basis to expand $x(t, \xi)$ [26]; that is,

$$x(t, \xi) = \sum_{j=1}^p c_j(t) \psi_j(\xi) \quad (19)$$

where $\{\psi_j(\xi)\}_{j=1}^p$ are the Hermite polynomials, p is the number of polynomials needed for the expansion, and $\{c_j(t)\}_{j=1}^p$ are the coefficients which need to be determined by CS method through step s3.

Once we have the PCE (19) for $x(t, \xi)$, we can recover the sparse coefficients in (19) by solving the following basis pursuit denoising (BPDN) problem:

$$\hat{c} = \arg \min \|c\|_1, \quad \text{subject to} \quad \|X - \Psi c\| \leq \epsilon \quad (20)$$

where X is the sample simulation results. If we use n as the number of sample simulation, then X is an n -dimensional vector at each time t , and Ψ is a $n \times p$ information matrix formed by inserting the stochastic sample points into Hermite polynomials. $c = \{c_j\}_{j=1}^p$ is the coefficient vector to be determined. For theoretical analysis regarding this method, readers can refer to [25].

B. Algorithm for the Stochastic Optimal Control Problem

To solve the optimal control problem (15), we need to introduce the Hamiltonian function H

$$H = \mathcal{J} + \lambda'(t) \{A(u)x(t) + Bu + C + DdW\} \quad (21)$$

where $\lambda(t)$ is system costate. Then, the equivalent Hamiltonian system for the optimal control problem is

$$dx = \frac{\partial H}{\partial \lambda}, \quad d\lambda = -\frac{\partial H}{\partial x}, \quad 0 = \frac{\partial H}{\partial u}. \quad (22)$$

To be specific, the corresponding system for control problem (15) with respect to (7) is given as

$$dx = A(u)x + Bu + C + DdW \quad (23)$$

$$d\lambda = \frac{-2\varepsilon_1 Q x e^{x'Qx}}{1 + e^{x'Qx}} - A(u)' \lambda - 2\varepsilon_3 a_1(a_1 x - b_1) \quad (24)$$

Algorithm 1: (CS-GDM Optimization Algorithm).

1. Initialize the optimization.
The optimization loop is initialized as follows.
 - 1) Select the number of Hermite polynomial basis p and the stochastic dimension d ,
 - 2) Select a step size parameter $z > 0$ and a tolerance parameter β ,
 - 3) Select a time step Δt ,
 - 4) Select an initial controller u^0 ,
 - 5) Using CS method to solve (23) to obtain the coefficients of expansion for system state x^0 ,
 - 6) Calculate the value $\mathcal{J}^0$ of the objective function (15),
2. Optimization loop.
For $l = 1, \dots$,
 - 1) From x^{l-1} and u^{l-1} , using CS method to determine the solution λ^{l-1} of the costate equation (24),
 - 2) From λ^{l-1} , u^{l-1} and x^{l-1} , determine the set of steps $\frac{\partial H}{\partial u^{l-1}}$ from (25),
 - 3) From u^{l-1} and $\frac{\partial H}{\partial u^{l-1}}$, determine the new value u^l

$$u^l = u^{l-1} - z^{l-1} \frac{\partial H}{\partial u^{l-1}},$$
 - 4) From u^l , using the CS method to calculate the coefficients of the expansion for system state x^l ,
 - 5) From u^l and x^l , determine the value of the functional J^l from (15),
 - 6) If $J^l \leq J^{l-1}$ and $(J^l - J^{l-1})/J^{l-1} \leq \beta$, stop and accept the result,
If $J^l \leq J^{l-1}$ and $(J^l - J^{l-1})/J^{l-1} \geq \beta$, go to step 1), incrementing the iteration counter l ,
If $J^l \geq J^{l-1}$, set $z^l = z^{l-1}/2$ and go to step 2).

$$0 = 2\varepsilon_2 Ru + 2\varepsilon_3 a_2(a_2 u - b_2) + \lambda \left(x \frac{\partial A(u)}{\partial u} + B \right). \quad (25)$$

Since system (23)–(25) is nonlinear, it is difficult to be solved directly. Thus, we tend to choose the iterative algorithm, such as gradient descent, conjugate direction, Newton, or quasi-Newton methods combining with CS method to solve this stochastic optimization problem numerically. In our proposed method, we choose gradient descent method as an example, and a new algorithm named as CS-GMD is proposed as Algorithm 1.

Instead of some conventional methods, e.g., the MC method, we use the CS method to calculate the system state $x(t)$ and the system costate $\lambda(t)$, which reduces computational cost dramatically and is more suitable for high-dimensional problems. This is regarded as one of the main contributions of this work. Now, we can numerically solve this complex stochastic optimal control problem. The obtained controller is indeed the control strategy for the management of energy storage in the considered ER. The relevant numerical results are given in Section V.

It is notable that our proposed CS-GDM algorithm can be used not only in the problem of ER energy management, but also in

TABLE I
VALUE OF PARAMETERS

Parameter	value	Parameter	value	Parameter	value
θ_l	0.4	μ_l	0.7	σ_l	0.4
θ_r	0.5	μ_r	0.2	σ_r	0.12
θ_p	0.1	μ_p	0.1	σ_p	0.43
T_{DG}	0.4	η	0.05	-	-
C_{max}	0.8	C_{min}	0.3	-	-

a variety of fields such as fluid mechanics, computational biology, and computational finance, where similar optimal control problems are expected to be solved.

V. SIMULATION RESULTS

In this section, the simulation results derived by our proposed method are given. First, we present some results to illustrate the feasibility of CS method in solving the stochastic differential equations. Then, the simulation results derived by our proposed CS-DGM algorithm for storage management problem of single MG and multiple interconnected MGs are given in the second part and third part separately.

All the solutions are obtained with the SPGL1 toolbox [27] in MATLAB (R2018b, Mathworks Inc., Natick, MA, USA) environment, and the hardware device is an Ubuntu 18.04 server with Intel Core i7-7700 CPU and one single GPU card with 2GB of graphic memory.

A. Simulation for State Equations

The main innovation of our proposed Algorithm 1 is to apply the CS method instead of other methods to deal with the stochastic state equation (7), and then combine it with optimization algorithm to solve optimization problems. So first of all, at this part, we verify the applicability and efficiency of CS method by comparing the simulation results with two traditional approaches, i.e., MC method and PCE method, for solving stochastic state equation (7). The values of all relevant parameters are given in Table I [5], [20].

Fig. 3 shows the sparsity of the coefficients c . We divide time interval $[0,1]$ into 100 subintervals, and the index is shown on the horizontal axis. The index of Hermite polynomial basis from (19) is shown on the vertical axis, which is from 0 to 20 in our numerical example. Each column in Fig. 3 corresponds to a coefficient vector c at different time nodes. To show the sparsity of coefficients c , only elements that are greater than a threshold τ are given as those dots. It can be seen that each component of $x(t, \xi)$ is sparse with the expansion of Hermite polynomials in stochastic space, which suggests the feasibility of CS method.

The effectiveness of the proposed CS method is demonstrated with Fig. 4 in which the power input/output of the energy storage ΔP integrated in the ER within 10 s is recovered from the solution to the BPDN problem (20) under the setting of $u_{ER} = 0.1$ and $u_G = 1$.

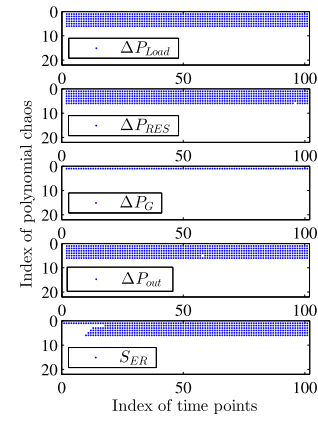


Fig. 3. Entries in coefficients c with absolute values larger than $\tau = 1 \times 10^{-5}$ within time interval $[0,1]$.

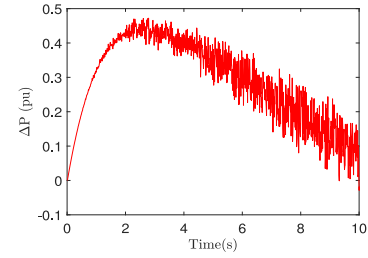


Fig. 4. Recovery of ΔP using CS method.

TABLE II
ERROR ESTIMATES FOR EXPECTATION RECOVERY FOR S_{ER} USING MC AND CS METHODS

Number of Sample(n)	Error ^{MC} $\times 10^{-2}$	Error ^{CS} $\times 10^{-2}$
10	7.83	12.71
20	5.49	2.20
30	4.69	0.72
50	3.35	0.006
100	2.41	0.006
150	2.05	0.006
200	1.69	0.006
1000	0.72	0.006

Next, to demonstrate the computational efficiency of the proposed method, we give the numerical error estimates for expectation recovery between the MC method and the CS method in Table II. It can be seen that when the same computation accuracy is achieved, 30 sample points are used via CS method, while 1000 sample points shall be used via the MC method. In addition, the CS method converges much faster than MC method, and increasing the number of samples will not provide additional information. Therefore, in real-world engineering practice, the CS method can save significant computational resource.

To further verify the accuracy of the CS method, the comparison results with the typical PCE method [18] are given in Table III. The main principle of the PCE method is to take expansion (19) into the state equations, and then determine the

TABLE III
ERROR ESTIMATES FOR S_{ER} USING CS AND PCE METHODS

	CS method	PCE method
Expectation error	6.1×10^{-5}	6.3×10^{-5}
Variance error	1.9×10^{-4}	2.3×10^{-4}
Sparsity(s)	0.2894	0.2579

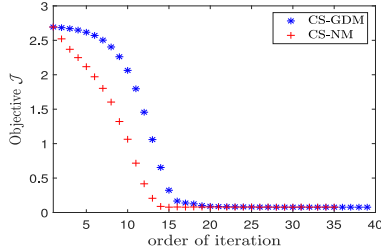


Fig. 5. Convergence of the objective function obtained by CS-GDM and CS-NM methods.

expansion coefficients $\mathbf{c}$ by solving a high-dimensional equations system (26).

$$d \sum_{j=1}^p c_j(t) \psi_j(\boldsymbol{\xi}) = \sum_{j=1}^p [A c_j(t) \psi_j(\boldsymbol{\xi}) + B u + C + D dW]. \quad (26)$$

The dimension of equation system (26) is p times of the original system (7), which makes the simulation of the PCE method more complicated. The error estimation of the expectation and variance for CS and PCE is obtained by comparing with 3000 sample points MC method. The sparsity is the ratio of the entries greater than τ in $\mathbf{c}$ to the total number of entries in $\mathbf{c}$. It can be seen that the CS method can achieve the same accuracy and sparsity with PCE method and does not need to rewrite the state equation solver, which is nevertheless necessary for PCE method.

In summary, the CS method not only overcomes the slow convergence of MC method, but also overcomes the complicated programming problem of PCE method, so it is more suitable to be combined with iterative method to solve stochastic optimization problems. In the following simulation parts, we verify the effectiveness of the algorithm1 in the scenarios of single MG and multiple interconnected MGs separately.

B. Simulation Results for Single MG

To demonstrate the solution to the considered control problem, we give the simulation results within time period $t \in [0, T]$, $T = 10$ s as an illustrative example. If the simulation results for a longer time scale is required, there is no essential difficulty to perform further simulations in different time periods. To be specific, in this case study, the weighting coefficients are set as $\varepsilon_1 = 1, \varepsilon_2 = 0.01, \varepsilon_3 = 1$. The initial values $x_0 = (0, 0, 0, 0, 0.5)$ and the initial control variables u_G and u_{ER} are random numbers from (0,1). The corresponding numerical results are given as follows. In Fig. 5, the blue spot is the value of objective function $\mathcal{J}$ derived by our Algorithm 1 at each step of the iteration, while the red spot represents the value of objective

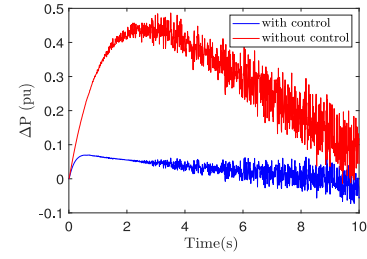


Fig. 6. Optimal ΔP with control vs. initial ΔP without control.

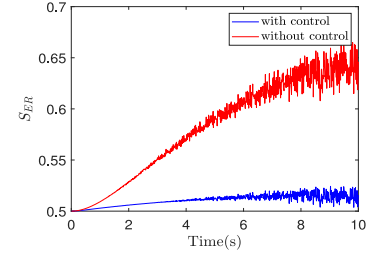


Fig. 7. Optimal S_{ER} with control vs. initial S_{ER} without control.

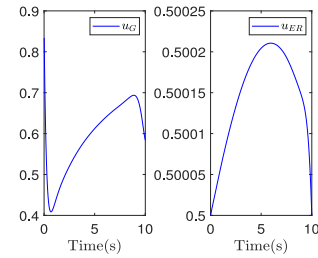


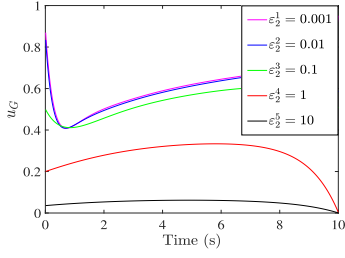
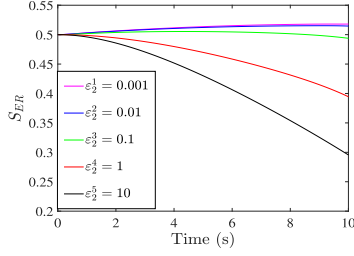
Fig. 8. Optimal controller u_G and u_{ER} .

function derived by combining CS with Newton method. They illustrate the effectiveness of combining CS method with iterative methods in solving the stochastic optimization problems. Although Newton method converges faster than our Algorithm 1, which is reasonable since Newton method has been proved to have higher convergence rate than gradient descent method for optimization problems, we still use gradient descent method as example to do simulation since it is easier to be coded.

Fig. 6 shows the behaviors of the energy storage in ER under two ER configurations, i.e., with/without active energy storage management. It can be seen that with the proposed control scheme, the power deviation of the considered energy storage in ER is significantly smaller than that when there is no active energy management scheme applied.

Similar to Fig. 6, Fig. 7 demonstrates the efficacy of the proposed optimal controller for the energy storage in ER. It can be seen that the proposed method is of the capability to achieve a better energy storage stability, which would be essentially important in most of practical EI scenarios.

The optimal controllers u_G and u_{ER} are given in Fig. 8. As expected, due to the penalty term (12), the optimal average transmission rejection rate u_{ER} for ERs has been properly kept within the constrained range. The optimal power adjustment u_G

Fig. 9. Range of u_G for different control weight ε_2 .Fig. 10. Expectation of optimal S_{ER} for different control weight ε_2 .TABLE V
COMPARISON OF THE ERROR BETWEEN MC AND CS METHODS

Optimal Objective	Err ^{MC} $\times 10^{-5}$	Err ^{CS} $\times 10^{-5}$
0.0106($\varepsilon_1 = 1, \varepsilon_2 = 0.01, \varepsilon_3 = 0.1$)	2.34	4.87
0.0776($\varepsilon_1 = 1, \varepsilon_2 = 0.01, \varepsilon_3 = 1$)	1.56	2.35
1.7280($\varepsilon_1 = 1, \varepsilon_2 = 1, \varepsilon_3 = 1$)	2.14	3.91

controls the power output of DGs with time to meet the stability of the system.

Next, to illustrate the impact of the emphasis for controller u_G on the optimal control scheme, the effectiveness of energy management schemes corresponding to different values of weight coefficients ε_2 for generators are evaluated. Here, we use the symbol $\varepsilon_2^i, i = 1, 2, \dots$ to represent for different ε_2 . Fig. 9 gives the optimal controller u_G for each ε_2 , which suggests that a smaller value of ε_2 would lead to a control signal u_G with larger variance. Therefore, in order to avoid overcontrol and ineffective control, we should properly choose the weight coefficients in the proposed optimal control problem (11).

Fig. 10 gives the expectation of energy storage level in the considered ER under a class of control schemes with different weight coefficient ε_2 . It indicates that when the cost for the adjustment of generators is relatively small, i.e., with small value of ε_2 , the energy storage level tends to be better stabilized.

Finally, we compare the time cost and accuracy for the MC method and our proposed CS-GDM algorithm by solving the same optimal control problem (15) with different sets of weight coefficients. In Table V, the error estimations are derived by comparing the results with 2000 points MC method. From Tables IV and V, it can be seen that when the accuracies are under the same level, CS-GDM method can improve the computational performance significantly. Apparently, the advantages of our

TABLE IV
COMPARISON OF THE COMPUTATIONAL COSTS BETWEEN MC AND CS METHODS

Optimal Objective	Time Cost ^{MC} (s)	Time Cost ^{CS} (s)
0.0106($\varepsilon_1 = 1, \varepsilon_2 = 0.01, \varepsilon_3 = 0.1$)	4672.84	423.91
0.0776($\varepsilon_1 = 1, \varepsilon_2 = 0.01, \varepsilon_3 = 1$)	3833.15	444.81
1.7280($\varepsilon_1 = 1, \varepsilon_2 = 1, \varepsilon_3 = 1$)	2934.98	278.64

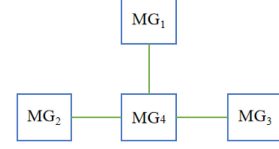
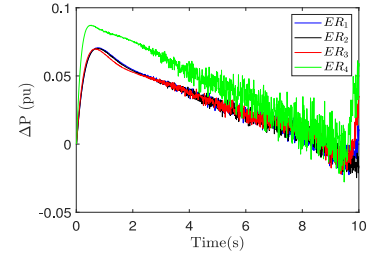


Fig. 11. Interconnected MGs topology for simulation

Fig. 12. Optimal ΔP for four ER_s .

proposed method in improving the efficiency of such numerical computation has been demonstrated.

C. Simulation Results for Interconnected MGs

In this subsection, the results for a more complex EI scenario are provided to demonstrate the efficacy of our proposed method. We consider four MGs in the network and they are functioning in the OFF-grid mode. The topology for this scenario is shown in Fig. 11. Without loss of generality, we assume that all MGs have the same components as is presented in the previous subsection, and the parameter settings are the same as before except for θ_p, μ_p in (5), which influence ΔP_{out} , the total power required from the other MGs. To be specific, let us set $\theta_p^1 = 0.1, \theta_p^2 = 0.2, \theta_p^3 = 0.5, \theta_p^4 = 0.8$ and $\mu_p^1 = 0.1, \mu_p^2 = 0.2, \mu_p^3 = 0.5, \mu_p^4 = 0.8$ for each MG. To ensure the balance of power transmission between MGs, constraint (27) needs to be added to the system. The weighting coefficients are set as $\varepsilon_1 = 1, \varepsilon_2 = 0.01, \varepsilon_3 = 1$. The corresponding numerical results are given as follows:

$$\sum_{i=1}^4 \Delta P_{out}^i = 0 \quad (27)$$

Fig. 12 shows the power dynamics of the energy storage of the four ERs with active energy storage management. It can be seen that with the proposed control scheme, the power deviation of the energy storage in ER_1, ER_2, ER_3 are almost the same, while more power needs to be stored in ER_4 than the others, since ER_4 needs to transfer power with the other ER_s .

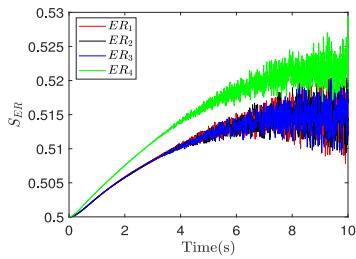


Fig. 13. Optimal S_{ER} for four ER_s .

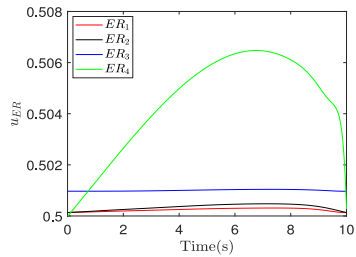


Fig. 14. Optimal controller u_{ER} for four ER_s .

TABLE VI

ERROR ESTIMATES FOR EXPECTATION RECOVERY FOR S_{ER} USING MC AND CS METHODS

Number of Sample(n)	Error ^{MC} $\times 10^{-2}$	Error ^{CS} $\times 10^{-2}$
30	10.45	23.59
50	8.18	10.42
80	6.46	4.21
100	5.29	1.12
500	2.31	0.054
1000	1.67	0.054
2000	1.39	0.054
3000	1.13	0.054

Similar to Fig. 12, Fig. 13 demonstrates the energy storage stability of the four ERs under control. It shows that the proposed method is of the capability to achieve a better energy storage stability. It also can be seen that the energy storage in ER_1 , ER_2 , ER_3 is more stable than that in ER_4 .

The optimal controllers u_{ER} are given in Fig. 14. Similar to results of the single MG, the optimal average transmission rejection rate u_{ER} for all the ERs have been properly kept within the constrained range. The adjustment of the transmission rejection rate of the ER_4 plays a major role in controlling the stability of this EI system.

At the end of this part, the comparison results for MC method and our CS-GDM method for the interconnected MGs are illustrated. First, Table VI shows the numerical error estimates for expectation recovery of S_{ER} for the interconnected state equations system. It can be seen that when the same computation accuracy is achieved, only 100 sample points are used via CS method, while 3000 sample points shall be used via the MC method. Therefore, the CS method can save significantly computational cost when solving the state equations. However, the key step to solve the control problem with gradient descent method is to solve the state equations iteratively. Therefore, it

TABLE VII

TIME COSTS BETWEEN MC AND CS METHODS VS THE NUMBER OF MGs

Number of MGs	Time Cost ^{MC} (s)	Time Cost ^{CS} (s)
1	4672.84	423.91
2	5463.32	607.11
3	7566.14	772.84
4	11426.35	901.76
5	12648.73	1240.51
6	15971.54	1426.23

is obvious that the CS-GDM method is much more efficient than the MC method to solve the same control problem. To be specific, as is shown in Table VII, for different number of MGs, the computational time cost is almost linear increases of both MC method and CS-GDM method, and MC method is nearly 10 times that of CS-GDM method. These comparison results fully show the advantages of CS-GDM method in computational efficiency.

To summarize, this section demonstrates the feasibility and effectiveness of our proposed CS-based stochastic optimal control method. Analyses of simulation results show the advantages of the CS-GDM approach we put forward in this work.

VI. CONCLUSION

In this article, an energy storage management problem for a typical ER was investigated. By modeling the MG system mathematically, this issue can be transformed into a nonlinear stochastic control problem. Then, by properly controlling desired power adjustment for DGs and the average transmission rejection for the studied ER, we can maintain the energy storage status at a stable level over time. The CS-GDM method was applied to solve such complex control problem. In the simulation part, the feasibility and validity of our proposed algorithm were verified. The comparison with the MC method in computational costs shows that our proposed algorithm can save computation time significantly, which means that such method is more suitable for complex and high-dimensional problems.

There are two main research directions in our future work: one way is to apply our algorithm to energy management problems for ERs considering game-theoretic multiobjective control problems. Another way is to combine the CS method with other existing methods such as ADP or reinforcement learning method to further improve the computation efficiency.

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learning algorithm.

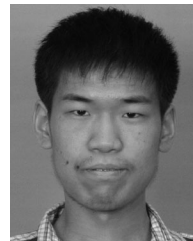
Hong Liang was born in Jilin, China, in 1991. She received the B.Sc. degree from Wuhan University, Wuhan, China, in 2013, and the Ph.D. degree from the Institute for Advanced Study, Tsinghua University, Beijing, China, in 2019, both in applied mathematics.

She is currently a Postdoctoral Researcher with the Department of Automation, Tsinghua University. Her research interests include control problems of energy internet, compressed sensing algorithm, and deep reinforcement



Haochen Hua (Member, IEEE) was born in Jiangsu, China, in 1988. He received the B.Sc. degree in mathematics and finance and the Ph.D. degree in mathematical sciences from the University of Liverpool, Liverpool, U.K., in 2011 and 2016, respectively.

From 2016 to 2020, he was a Postdoctoral Fellow with the Research Institute of Information Technology, Tsinghua University, Beijing, China. Since 2020, he has been a Professor with the College of Energy and Electrical Engineering, Hohai University, Nanjing, China. He has authored or coauthored more than 50 papers and has authored three books. His research interests include energy Internet system modeling and optimization, optimal and robust control theory, and stochastic calculus.



reinforcement learning and their applications in healthcare.

Yuchao Qin was born in Henan, China, in 1994. He received the B.Sc. degree in automation and the M.S. degree in control science and engineering from Tsinghua University, Beijing, China, in 2017 and 2020, respectively. He is currently working toward the postgraduation degree with the Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, U.K.

His research interests include control and optimization, reinforcement learning, and inverse



Maojiao Ye (Member, IEEE) received the B.Eng. degree in automation from the University of Electronic Science and Technology of China, Sichuan, China, in 2012, and the Ph.D. degree in theory and control engineering from Nanyang Technological University, Singapore, in 2016.

She is currently a Professor with the School of Automaton, Nanjing University of Science and Technology, Nanjing, China. From 2016 to 2017, she was a Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University. Her research interests include game theory and distributed optimization and their applications.

Dr. Ye was the recipient of the Guan Zhao-Zhi Award in the 36th Chinese Control Conference 2017 (first author) and the Best Paper Award in the 15th IEEE International Conference on Control and Automation 2019 (sole Author).



Shuqing Zhang (Member, IEEE) was born in 1983. He received the B.Sc. and Ph.D. degrees in electrical engineering from Tsinghua University, Beijing, China, in 2005 and 2010, respectively.

From 2010 to 2012, he was a Postdoctoral Fellow with the Department of Electrical Engineering, Tsinghua University, where he has been an Assistant Professor since 2012. His research interests include hybrid real-time simulation of large HV ac/dc power grid, key techniques and simulation of Internet of Things, dynamic simulation of multienergy system and energy Internet, and power load measurement-based electric load identification and modeling.



Junwei Cao (Senior Member, IEEE) received the bachelor's and master's degrees in control theories and engineering from Tsinghua University, Beijing, China, in 1996 and 1998, respectively, and the Ph.D. degree in computer science from the University of Warwick, Coventry, U.K., in 2001.

He is currently a Professor with Beijing National Research Center for Information Science and Technology, Tsinghua University. He has authored or coauthored more than 200 papers and cited by international scholars for more than 18000 times. He has authored or edited eight books. His research interests include distributed computing technologies and energy or power applications.

Dr. Cao is a Senior Member of the IEEE Computer Society and a Member of the ACM and CCF.